

Difficulty Estimation of Rush Hour

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¹*Scientific Perspectives on GMT, Utrecht University, 19 September 2024*

In this paper, we propose a weighted linear difficulty function, as well as a weighted nonlinear difficulty function, to measure the difficulty of levels in the puzzle game Rush Hour. To this end, we build on top of the ideas presented in the paper of van Kreveld et al., identifying game features. We found the nonlinear difficulty function to be able to capture aspects of game-level difficulty that are otherwise not measurable using a purely linear difficulty function.

I. INTRODUCTION

Estimating the difficulty of games is a seemingly simple task that can be surprisingly complex. Puzzle games serve as a good platform to analyze difficulty in games because of their rigid nature; in most puzzle games, there is an initial state (i.e., initial configuration) and one or multiple solution states (i.e., solved configurations) for each level. These levels can be broken down into a series of game features (i.e., game properties) that each affect the difficulty perceived by the player, to various extents and in different ways.

These features can be classified as *initial features*, *solution features*, and *dynamic features*. Initial features can be extracted from the initial state of a level, solution features can be extracted from the solution state of a level, and dynamic features are related to the solving process of a level [1]. We can use these features to design *measures* (i.e., any function from ‘something’ to the nonnegative extended real numbers) for puzzle games, seeking to quantify certain aspects of those games. Although puzzle games are relatively well-suited for quantification and therefore difficulty analysis, designing good measures that correlate strongly to the difficulty of a puzzle game remains nontrivial.

In this paper, we analyze a generic version of the puzzle game Rush Hour to design two difficulty functions that each map features of the game to a difficulty score. We first select features that we expect to correlate with the difficulty of a Rush Hour level, and then use the selected features in a weighted linear difficulty function. We then show various combined measures that could be used in a weighted nonlinear difficulty function. The nonlinear function is aimed to improve the linear function by introducing methods that we expect to correlate better with the actual difficulty of a Rush Hour level.

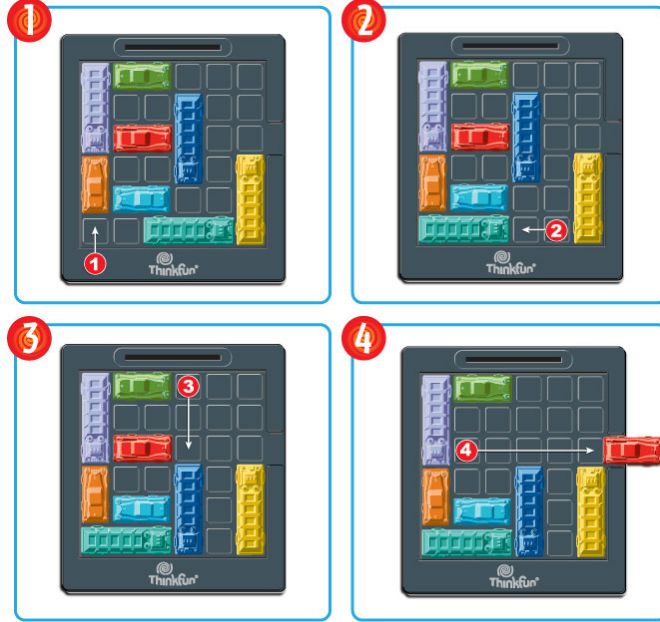


FIG. 1. Rush Hour board setup (1) and steps of the solution path (2-4). Image source: Meeples, "Rush Hour," 2024. Accessed via <https://meeplesbordspellen.nl/en/product/rush-hour-2/>.

II. PUZZLE GAME DESCRIPTION

Rush Hour is a puzzle board game that centers around getting the player’s car out of a traffic jam. The traffic jam is represented on a game board (with a default size of 6×6 tiles) using a series of cross-directional obstacles in the form of sedans and trucks. Sedans are defined as two-tile obstacles, and trucks are three tiles wide. The cars (both sedans and trucks) are only allowed to move forward or backward in the direction of their bodies. When the player’s car leaves the exit, it means the level is cleared. In the generic version we analyze in this paper, the grid size is a dynamic variable that can be adjusted for added difficulty, allowing us to analyze a wider range of level possibilities.

III. GAME FEATURES THAT INFLUENCE DIFFICULTY

We describe multiple game features of Rush Hour that we identified to possibly influence the difficulty of a level, as experienced by the player. To approximate and quantify these features, we have come up with the following measures, classified into three categories: initial features, solution features, and dynamic features, as defined in van Kreveld et al. [1].

A. Initial features

- *Grid size.* The initial implementation of Rush Hour has a constant 6×6 grid of tiles. In the generic representation of Rush Hour, a dynamic board size would be one of the most important factors in determining the difficulty of the puzzle, due to the quickly growing complexity of having more possible moves as the grid size increases.
- *Number of sedans.* The two-tile-wide sedan is one of the car types that form an obstacle the player must deal with. A larger number of sedans in a level results in more interactable objects, more visual clutter (i.e., visual information that the player needs to process), and expectedly less freedom of movement for other cars. The number of sedans therefore seems to be a good measure of the difficulty of a level. We expect the small size of a sedan to generally afford the player the option to make adjustments to the greater puzzle more easily than a truck would.
- *Number of trucks.* The three-tile-wide truck is the other car type and takes up half of a row or column in the default grid. The large size of the trucks makes them relatively cumbersome to clear out of the player’s path, compared to the sedans. We expect the number of trucks to be a good measure of the difficulty of a level, for the same reasons as provided for the number of sedans. The number of trucks might correlate more strongly with the difficulty than the number of sedans would, as the comparatively large size of trucks could impose a stronger restriction on the overall freedom of movement within a level.
- *Number of perpendicular cars.* In order to complete the level, the player’s car must always move in the direction of the static exit. The number of perpendicular cars is defined as the number of cars, regardless of type, that are oriented along the grid axis that is perpendicular to the exit direction of the level. Increasing the number of perpendicular obstacles the player must move adds complexity to the desired solution path, as opposed to a board filled with only parallel obstacles.

B. Solution features

- *Number of moves in optimal solution path.* Rush Hour levels generally seem to be less difficult when a small number of moves is required to reach the solution. A level with a small number of moves in the optimal solution path typically would involve moving a few obstacles out of the way to form a clear path for the player. On the contrary, levels that are filled with relatively many obstacles that need to be shifted around the grid in the solution path tend to be more difficult.

C. Dynamic features

- *Number of counterintuitive moves in the optimal solution.* Counterintuitive moves are defined as moves in an optimal solution path that move the player’s car further away from the exit. Counterintuitive moves are expected to be harder for the player to find, especially when multiple counterintuitive moves are chained in a row. The number of counterintuitive quantifies the difficulty of the moves that are needed to solve the puzzle optimally.

The above selection of features results in the following weighted linear difficulty function:

$$\text{difficulty} = W_1 \cdot \text{sedans} + W_2 \cdot \text{trucks} + W_3 \cdot \text{relevantCars} + W_4 \cdot \text{gridSize} + \\ W_5 \cdot \text{minimumMoves} + W_6 \cdot \text{counterIntuitiveMoves} + W_7 \cdot \text{perpendicularCars}$$

IV. NONLINEAR OPTIONS FOR A DIFFICULTY FUNCTION

The weighted linear difficulty function we designed is a basic function that might do decently well at estimating the difficulty of a Rush Hour level. However, some aspects of Rush Hour (e.g., interactions between game features) that could contribute to the difficulty of a level might be impossible to express as a linear combination of game features. To account for this, we introduce nonlinear terms in our linear function, in the form of combined measures. We describe why these nonlinear terms might improve on, or add to, the linear difficulty function.

- *Ratio of trucks to sedans.* As explained, a truck takes up three tiles, compared to the two tiles of a sedan. We expect that a balance between the number of trucks and the number of sedans decreases the overall freedom of movement within a level more than a high or low ratio of either car type. We denote this balance as *car variety*. When there is an equal number of trucks and sedans in the level, then the car variety is the highest. When there is only one car type present in the level, the car variety is the lowest. This is expressed through the combined measure:

$$\text{carVariety} = \frac{\text{trucks}}{\text{totalCars}} \cdot \frac{\text{sedans}}{\text{totalCars}} = \frac{\text{trucks}}{\text{totalCars}} \cdot \left(1 - \frac{\text{trucks}}{\text{totalCars}}\right)$$

In turn, we expect a correlation between the lack of freedom of movement (i.e., movement restriction), imposed by the car variety, and the difficulty of a level. Because of this, we opt to include the ratio of trucks to sedans in our non-linear difficulty function. We think this combined measure captures an important aspect of the degree of movement restriction caused by the car types that are present in the level.

- *Ratio of cars area to grid size.* As explained before, there might be a correlation between the freedom of movement and the difficulty of a level. A factor that we expect to influence freedom of movement is the occupied area relative to the total area available, which we will call the *board flexibility*. The board flexibility would be the lowest when the board is completely occupied, or completely unoccupied. The board flexibility would be the highest when the grid is exactly half-occupied. The board flexibility is expressed through the combined measure:

$$\text{boardFlexibility} = \frac{\text{carsArea}}{\text{gridSize}} \cdot \frac{\text{gridSize} - \text{carsArea}}{\text{gridSize}} = \frac{\text{carsArea}}{\text{gridSize}} \cdot \left(1 - \frac{\text{carsArea}}{\text{gridSize}}\right)$$

The board flexibility quantifies a balance between the number of movable pieces and the available space for those pieces to move. The area occupied by cars is not one of our selected game features, but can be calculated from them.

- *Ratio of exit-direction cars to perpendicular-direction cars.* If we treat a car on the board as a line along its body direction, then the difficulty of a level is expected to increase when these lines intersect more frequently. The degree of intersection of these lines can be described by *road complexity*. Road complexity is determined by the number of cars moving along the exit direction (i.e., the body direction of the player's car) and the number of cars moving along the perpendicular direction. When the number of cars facing the exit direction is equal to the number of cars facing the perpendicular direction, then the road complexity is the highest. When the number of any type of car is zero, the road complexity is the lowest. This is expressed through the combined measure:

$$\text{roadComplexity} = \frac{\text{exitCars}}{\text{totalCars}} \cdot \frac{\text{perpendicularCars}}{\text{totalCars}} = \frac{\text{perpendicularCars}}{\text{totalCars}} \cdot \left(1 - \frac{\text{perpendicularCars}}{\text{totalCars}}\right)$$

We expect the road complexity to correlate with the difficulty of a level, as cars that have intersecting movement lines might block each other more easily.

- *Ratio of relevant cars to all cars.* Relevant cars are defined as cars that need to be moved in any optimal solution path of the level. By looking at the ratio of relevant cars to all cars in the level, we can approximate

how difficult it is to find the right moves from all options available to the player. Cars that are not needed to solve the puzzle only add to the visual clutter of the game (which is already accounted for in other measures). With this measure, we want to put a higher emphasis on cars that matter to the solution path.

The combined measures described above incorporated into our weighted linear difficulty function results in a weighted nonlinear difficulty function. Since we make use of the same function for multiple of the combined measures, we will first define that function separately for clarity. Let $f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be defined by:

$$f(x, y) = \frac{x}{y} \cdot \left(1 - \frac{x}{y}\right)$$

Our nonlinear formula then becomes as follows:

$$\begin{aligned} difficulty = & W_1 \cdot sedans + W_2 \cdot trucks + W_3 \cdot gridSize + \\ & W_4 \cdot minimumMoves + W_5 \cdot counterIntuitiveMoves + W_6 \cdot f(trucks, totalCars) + \\ & W_7 \cdot f(carsArea, gridSize) + W_8 \cdot f(perpendicularCars, totalCars) \end{aligned}$$

V. FURTHER RESEARCH

The nonlinear ratios are a powerful tool to analyze game features otherwise not possible to quantify using linear measures. While it was not within the scope of this study, there are a few steps needed to be able to use our designed difficulty functions. This involves calculating the weights for the terms corresponding to the selected measures as per Section III. It could also be interesting to conduct a study to research the correlation between each of the designed measures with the difficulty perception of players. and see whether the non-linear measures correlate equally or better with the true difficulty of a level in comparison with the linear measures. Other avenues for further research would include designing more measures, aimed to capture possibly meaningful aspects of Rush Hour related to difficulty. Lastly, testing the effectiveness of nonlinear measures in other puzzle games could be an interesting topic of study.

Hours: Xiao Yi Hu: 13:25h Binxu Jiang: 13h Brendan Miller: 13h Jens Steenmetz: 17h

Task division: Everyone worked together during in-person meetings writing up sections distributed evenly and we conducted post-writing reviews to perfect the paper and formatting.

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- [1] Marc van Kreveld, Maarten Löffler, and Paul Mutser. Automated puzzle difficulty estimation. In *2015 IEEE Conference on Computational Intelligence and Games (CIG)*, pages 415–422, 2015. doi:10.1109/CIG.2015.7317913.